AI for Transportation: From concepts to implementation

Lecture 3: Case Studies

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MIT Open Learning

JTL | MIT TransitLab

AI for Transportation

L1: Introduction

L2: AI Models and Approaches

L3: Real-World AI Case Studies

L4: Generative AI

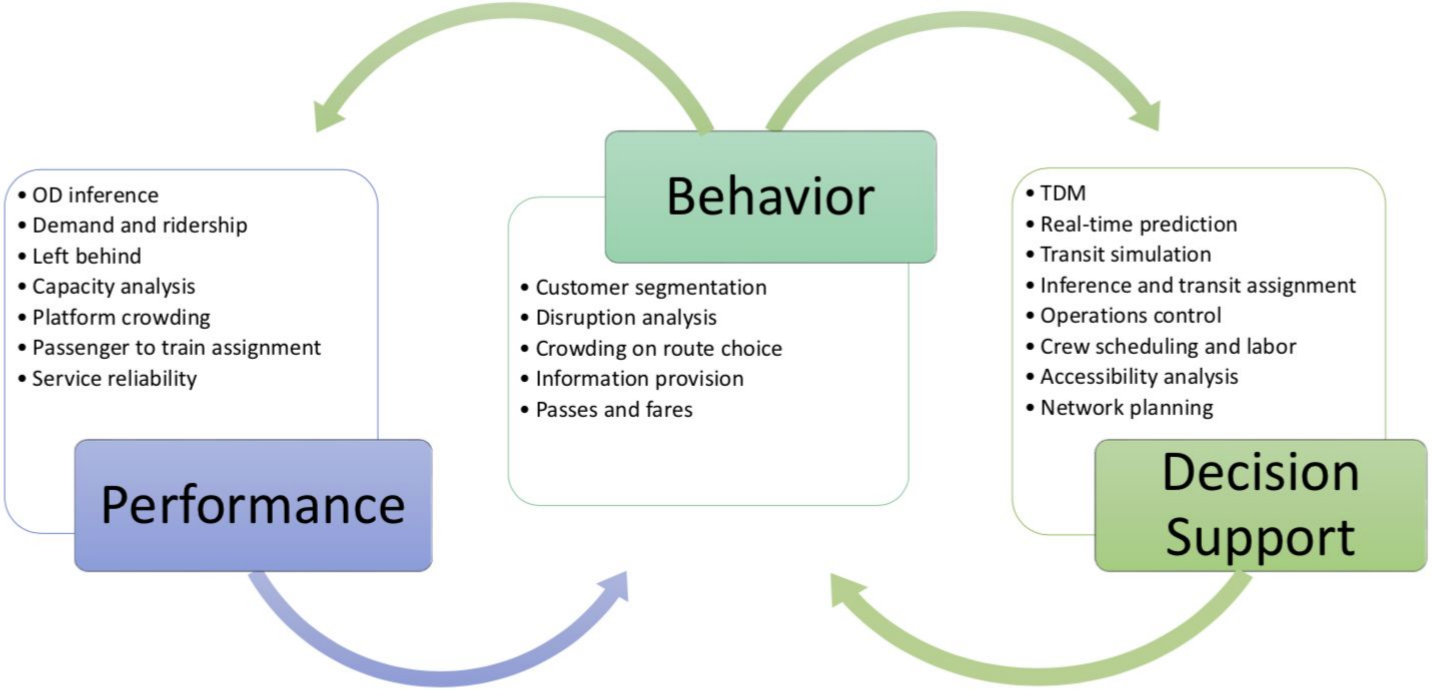
MIT Global Partnerships

Transport for London (TfL)	Chicago Transit Authority (CTA)
2005~2021	2001~2010, 2016~Present
Hong Kong (MTR)	Boston (MBTA)
2012~2019	2003~2005, 2010~Present
NRF, Singapore (SMART)	Washington DC (WMATA)
2016~2021	2021~Present

Cross-Cultural Learning

London, Chicago, Boston, D.C., Hong Kong, Shanghai, and Singapore

Research Areas



Application Levels

Operational

- Operational control
- Disruption Recovery
- Crew scheduling and labor
- Service Planning and Scheduling

Tactical

- Behavior / choice
- Passes and Fare Policy
- OD inference
- Demand and ridership
- Travel Demand Management

Strategic

- Long-term planning
- Funding and financing
- Network planning
- Accessibility and Wider Benefit Assessment

Clusters — Applied Data Analytics Projects

Customer Behavior & Predictive Analytics

1. Multimodal OD Estimation: Oyster+
2. Reliability: Monitoring, Improving and Informing
3. Predictive Analytics in London Underground
4. Travel Pattern Recognition based on Oyster

Decision Support & Performance Measurement

5. Disruption Management & Decision Support for London Underground
6. Bus Operation under Road Construction
7. Bus Network Design / Improvement

Planning/Policy

8. Transit Connectivity: Communication and Capacity
9. Autonomous Vehicles and Public Transit Integration
10. Demand Responsive Transit + Network Improvement

AI for Transportation

L3: Case Studies

S1: AI to Detect Platform Crowding

S2: AI for Bus & Train Dispatching Control

S3: AI to Understand Bus Operators' Preferences

Three case studies

Predict

AI to Detect Platform Crowding

Riccardo Fiorista, Awad Abdelhalim & Anson Stewart with WMATA

Sense **Assist**

Control

AI for Bus & Train Dispatching Control

Joseph Rodriguez, Haris Koutsopoulos & Jinhua Zhao with CTA

Assist

Represent

AI for Sentiment Analysis and Complaint Categorization

Michael Leong & Awad Abdelhalim with WMATA

Sense

Predict | Sense | Assist

AI to Detect and Predict Platform Crowding

Riccardo Fiorista, Awad Abdelhalim & Anson Stewart

Near Capacity Operation

Crowd Management

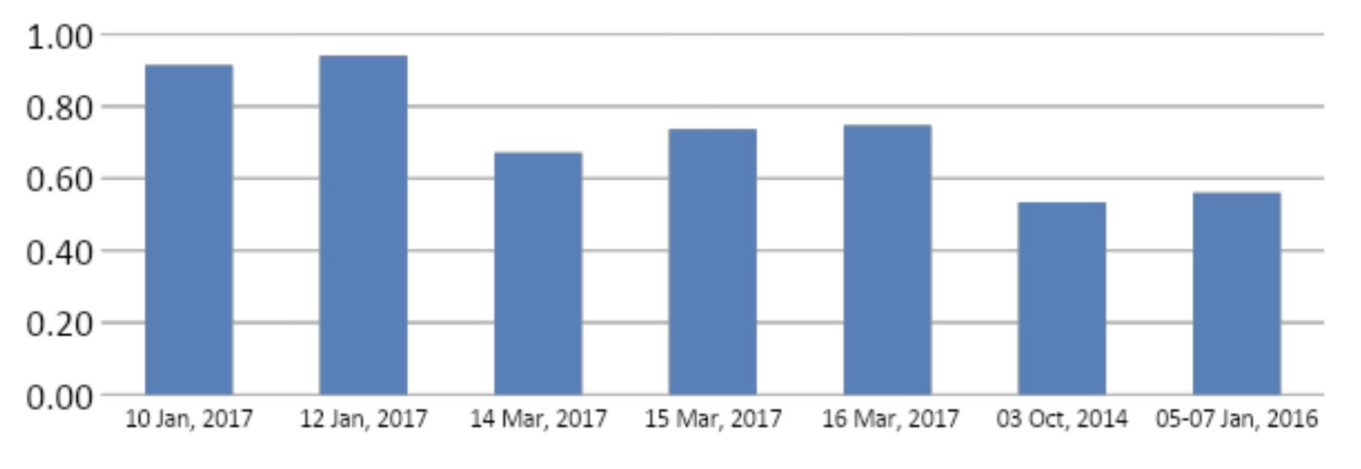
Inference (Performance)	Prediction (Real-time)	Forecast (Planning)
• Denied boarding • Path choice • Performance monitoring	• User segmentation • Crowding prediction • Demand prediction	• TDM (promotion) • Operating strategies • Planning
UNDERSTAND	**INFORM**	**IMPROVE**

	Analysis	Customer	Operation
Recurring	Left Behind	• Information • TDM	Decision Support
Special	Incidence	Information	Decision Support

Near Capacity Operation

Left Behind Rate Validation

% of passengers cannot boarding the first train



Admiralty Station, Tsuen Wan Line UP (18:00-19:00)

Automated Left Behind Estimation

DATABASE
AFC, AVL

ESTIMATION UNIT

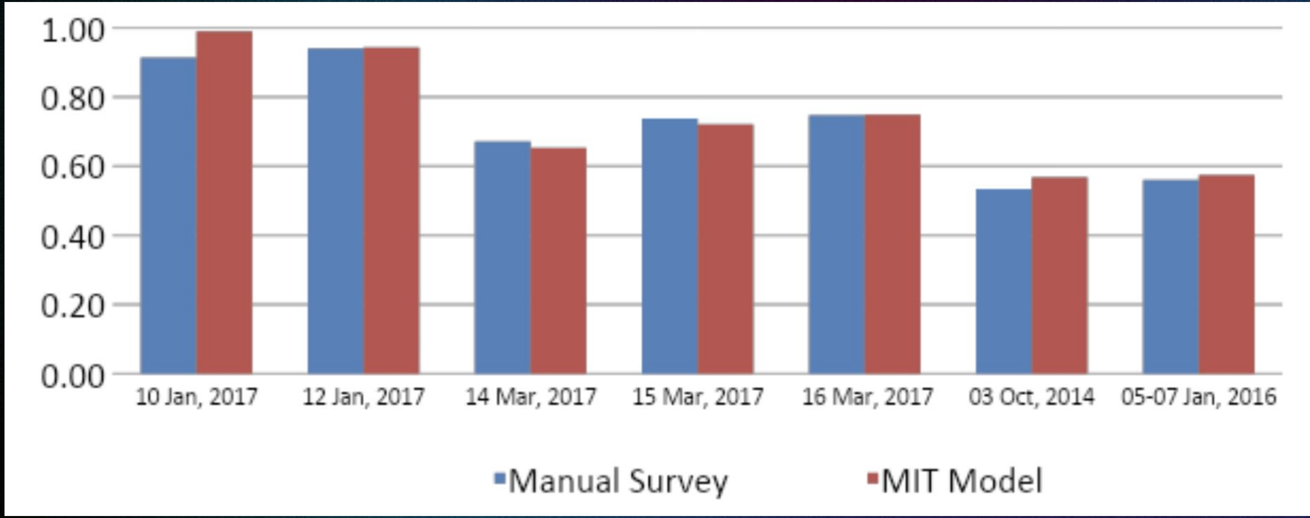
Data Preparation Module

Left Behind Estimation Module

USER INPUTS
- *Station*
- *Time period*

Left Behind Rate Validation

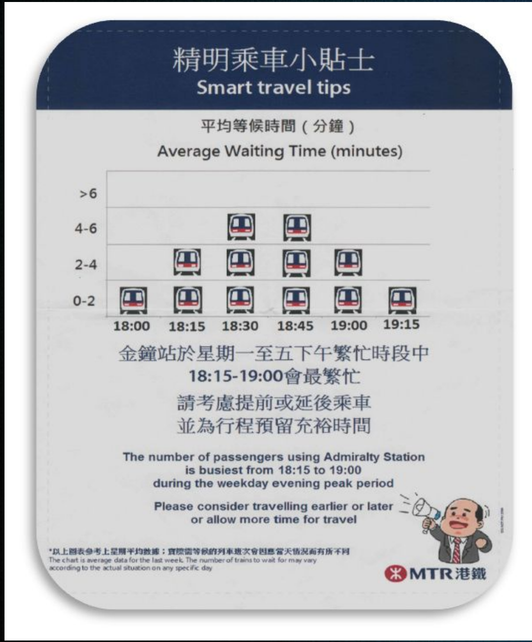
% of passengers cannot boarding the first train



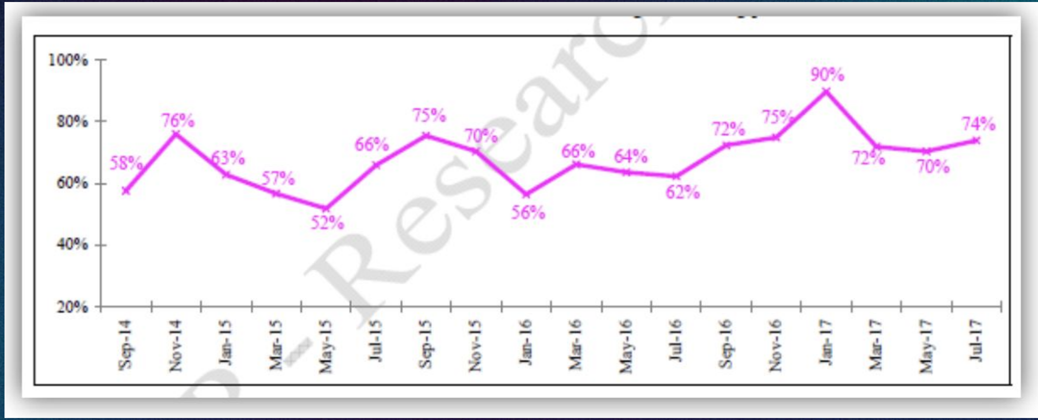
ADM TWL UP (18:00-19:00)

Customer Information and Government Report

Customer: Travel Tips



Government Report



Why is platform crowding important?

Safety

- On platforms / in stations for **customers** and **staff**

Operational Efficiency

- **Anticipate delays** and necessary **communication** with customers and emergency services

Customer Experience

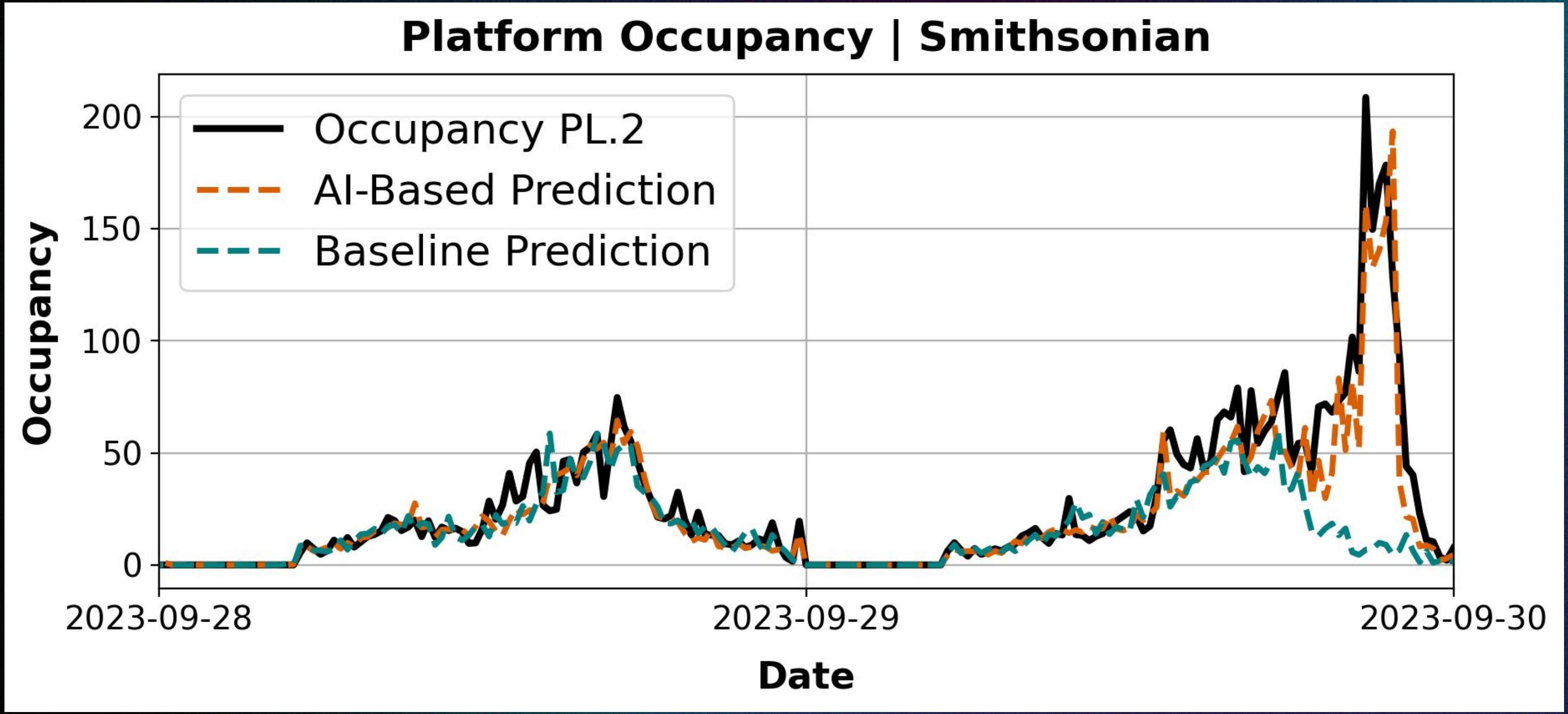
- Provide better **passenger information** and allow for **better journey planning**

Robust models allow combination of data sources

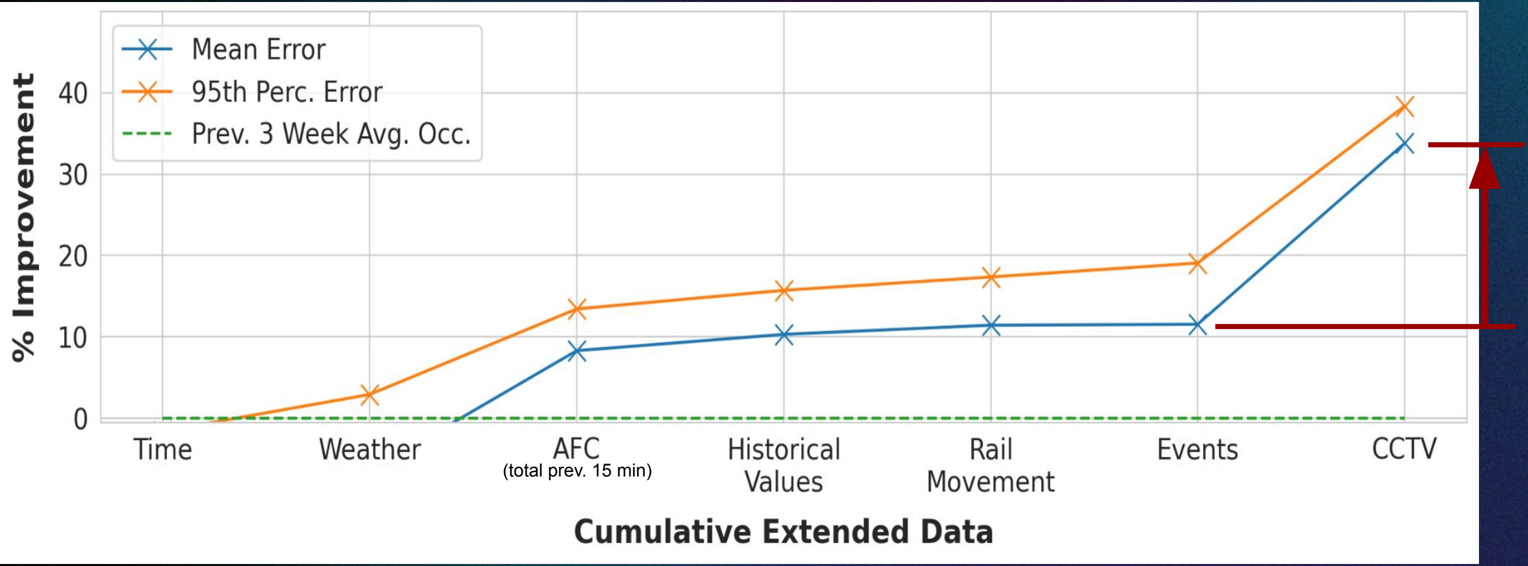
- Target
- Historic
- Real-Time
- Future

	Historic	Real-Time	Future
ODX Estimated Platform Occupancy (Target)	✓		
Automatic Vehicle Location (AVL)	✓	✓	
Scheduled Rail Movement	✓	✓	✓
Automatic Fare Collection (AFC)	✓	✓	
Weather	✓	✓	✓
Planned Events	✓	✓	✓
Revenue Service Adjustments	✓	✓	✓
CCTV Data	✓	✓	

Innovation #1: Operational data + AI

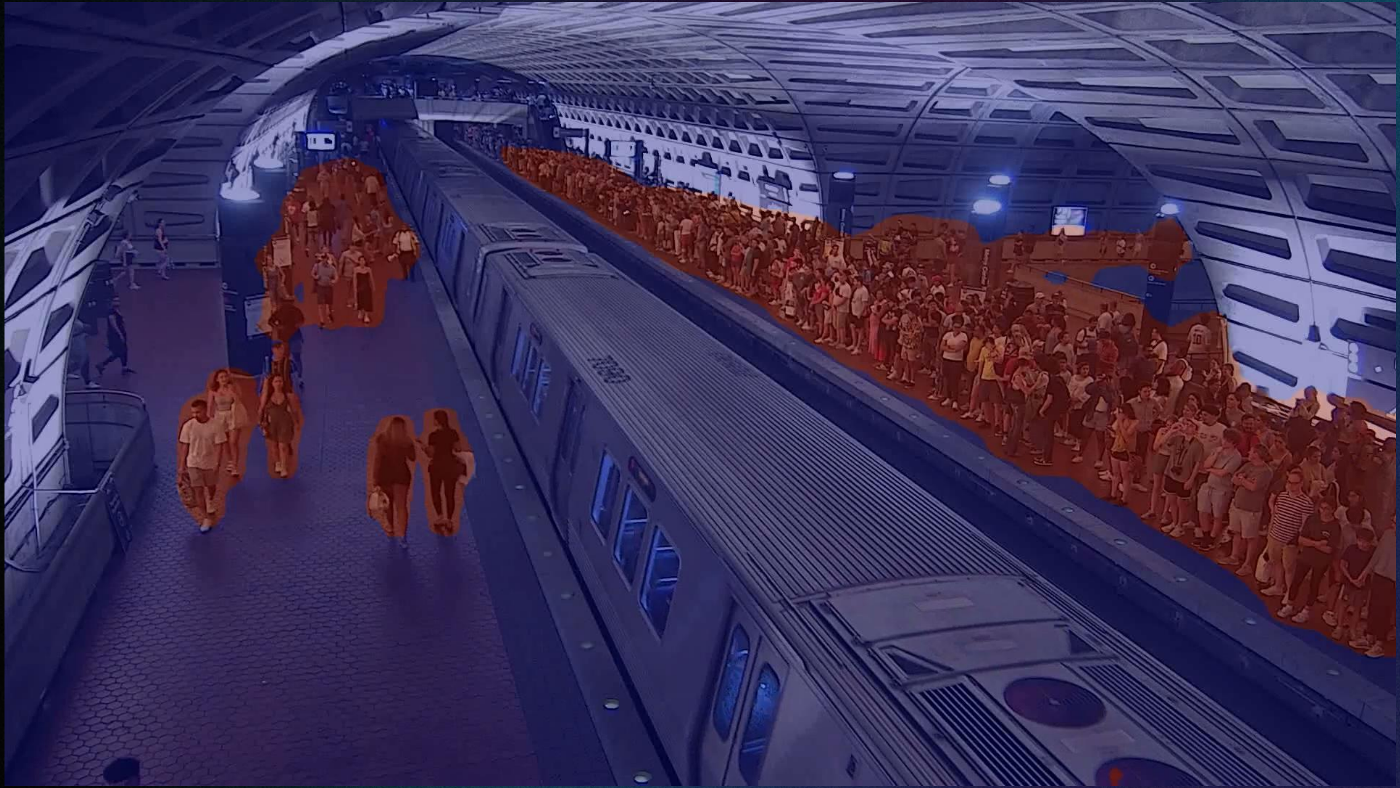


Innovation #2 Operational data + AI + real-time CCTV

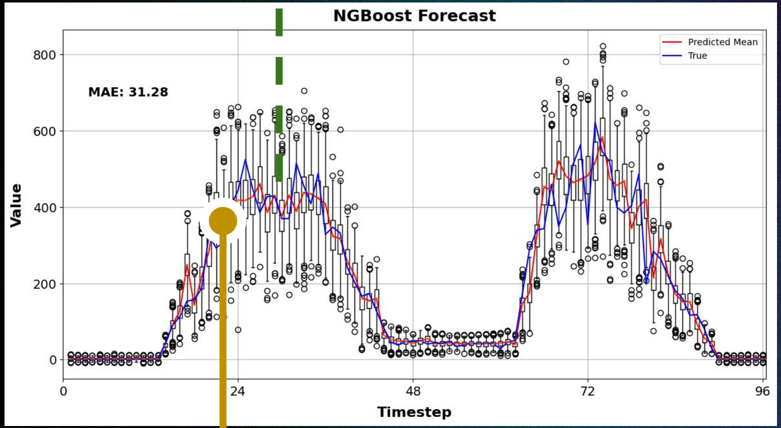


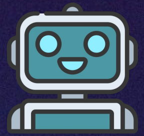
Improvement over best baseline: > 30%

Image segmentation provides an **accurate proxy** of platform crowding levels with **low data requirements**



Innovation #3: Use CCTV data only when uncertainty is high





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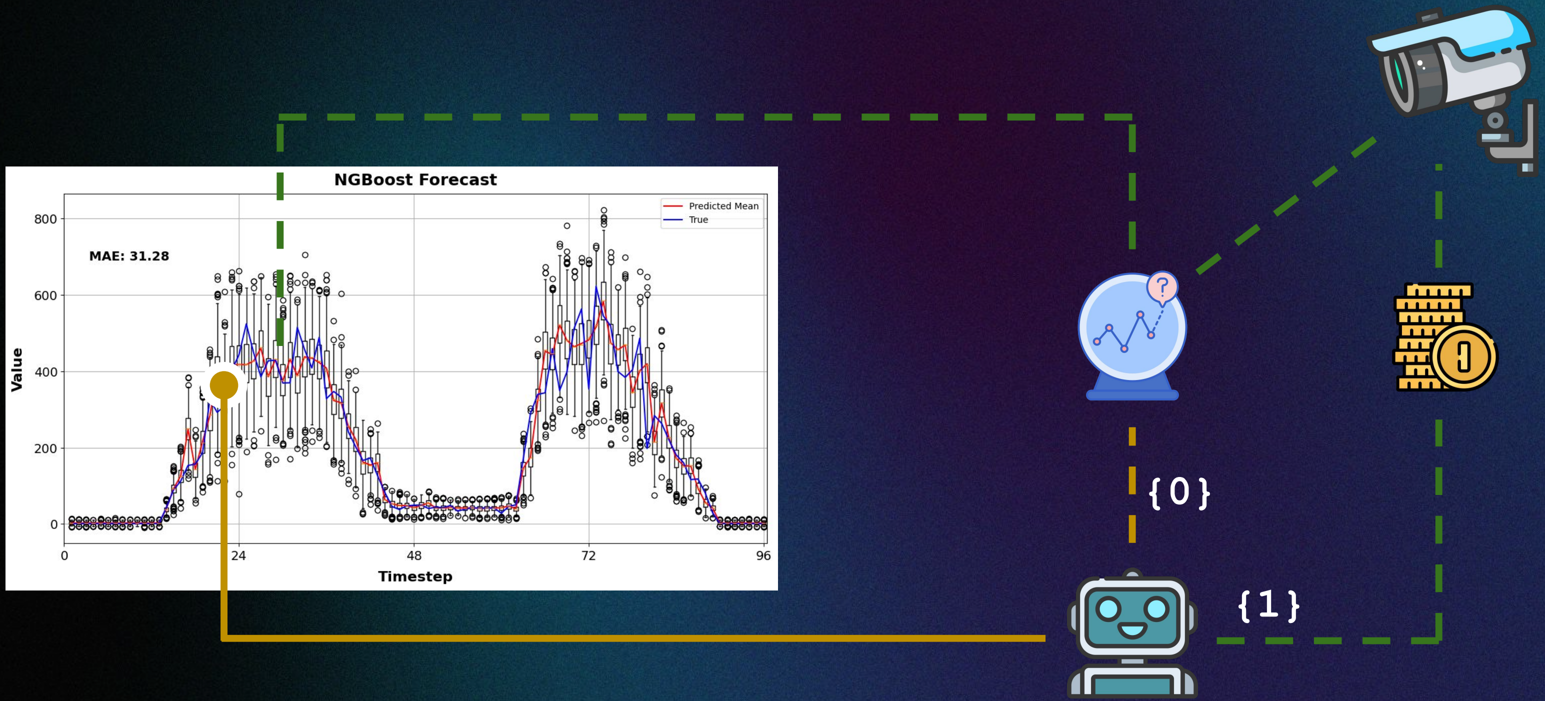


Observation #1 - Ethical use of AI

Observation #2 - How do we use history?

How do we use history?

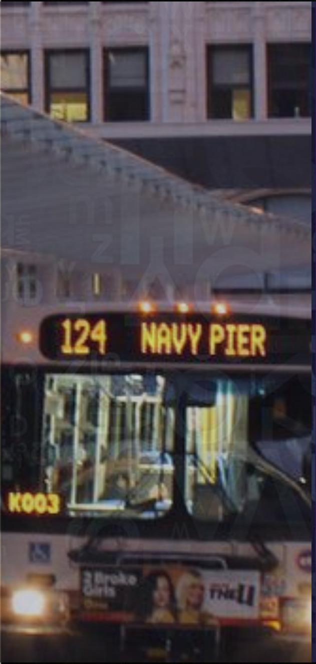
Amazing how history repeats itself

Historic mean is often very good

History that is most relevant today

Component vs. Whole of History

De-compose and re-mix



Control

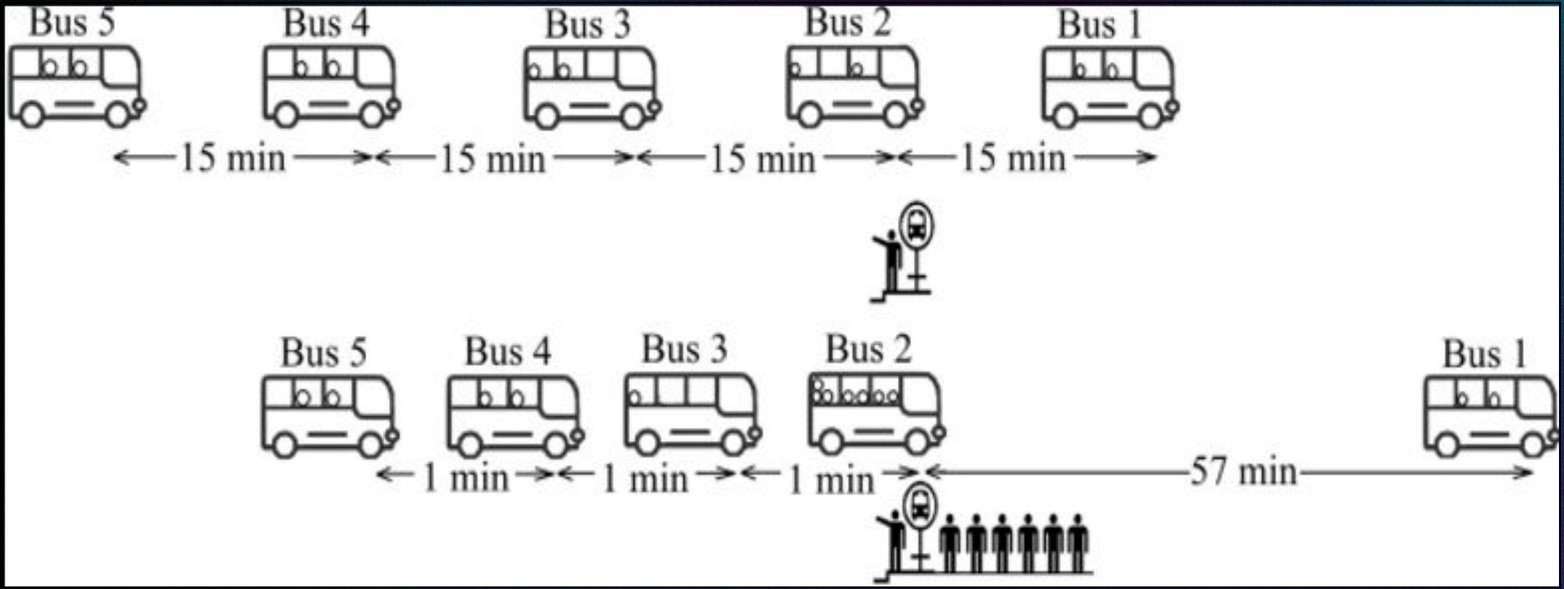
Assist

AI for Bus Dispatching to Improve Reliability

Joseph Rodriguez, Haris Koutsopoulos & Jinhua Zhao


1 HARVARD

Bus bunching



Control strategy

At the bus terminal,

1. Hold the bus by 1 min, 2 mins, 3mins, ...
2. Move departure time forward by 1 min, 2 mins, 3 mins,...

Transit agencies experience major service reliability issues caused by driver shortage and driver absenteeism

CTA Service delivered: 84%

Transit agencies experience major service reliability issues caused by driver shortage and driver absenteeism


TransitCenter

Bus Operators in Crisis

The Steady Deterioration of One of Transit's Most Essential Jobs, and How Agencies Can Turn Things Around

CTA Service delivered: 84%

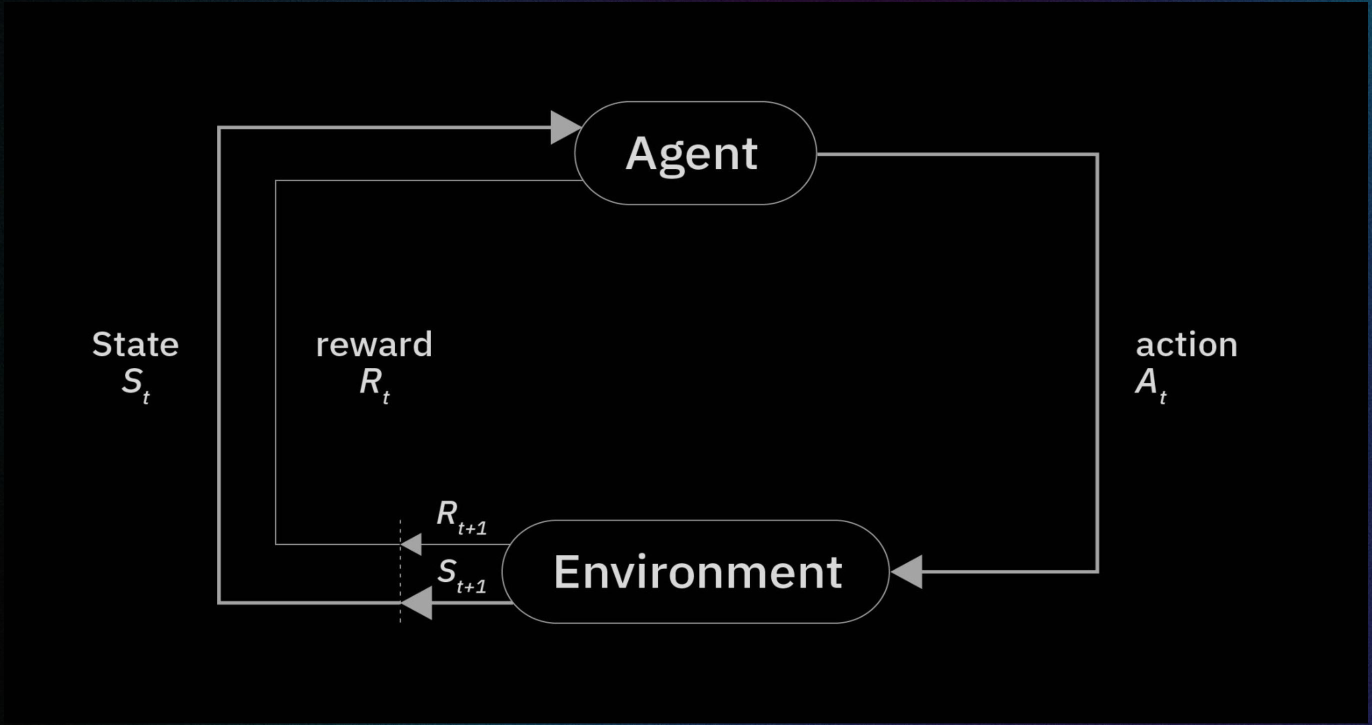
Improve service reliability through bus control using reinforcement learning

Robustness to uncertainties:

- Demand uncertainty: # of passengers
- Supply uncertainty: driver absenteeism

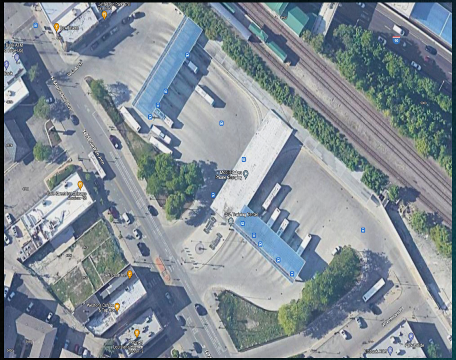
- Environment uncertainty: traffic conditions
- Operation uncertainty: non-compliance
- Information uncertainty: Incomplete or inaccurate

Reinforcement Learning

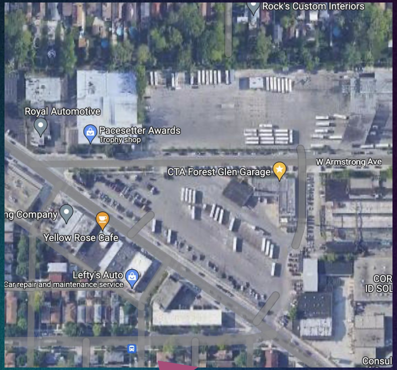


Human in the loop #1: Bus route 81 (actual) vs. 20 (initial)

Terminal: Jefferson Park, serves 10 routes

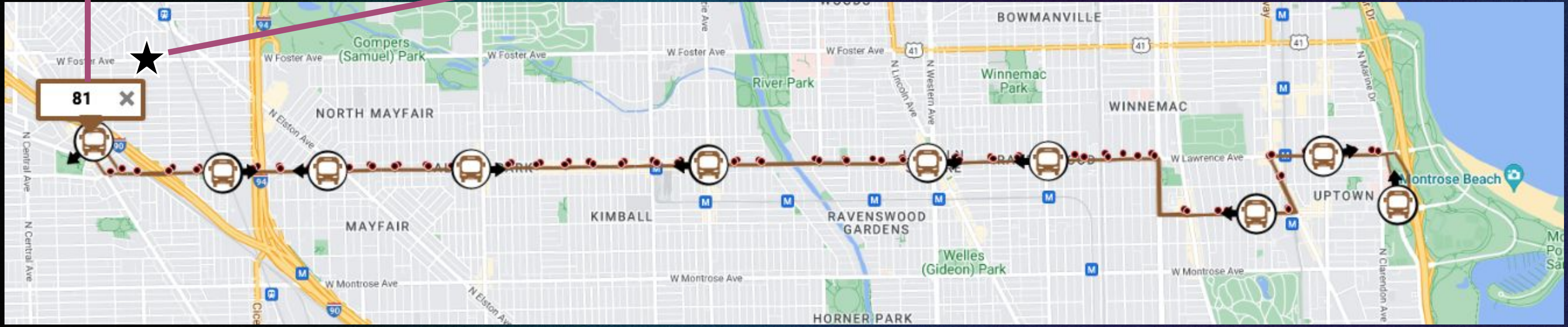


Garage: Forest Glen, North Chicago routes

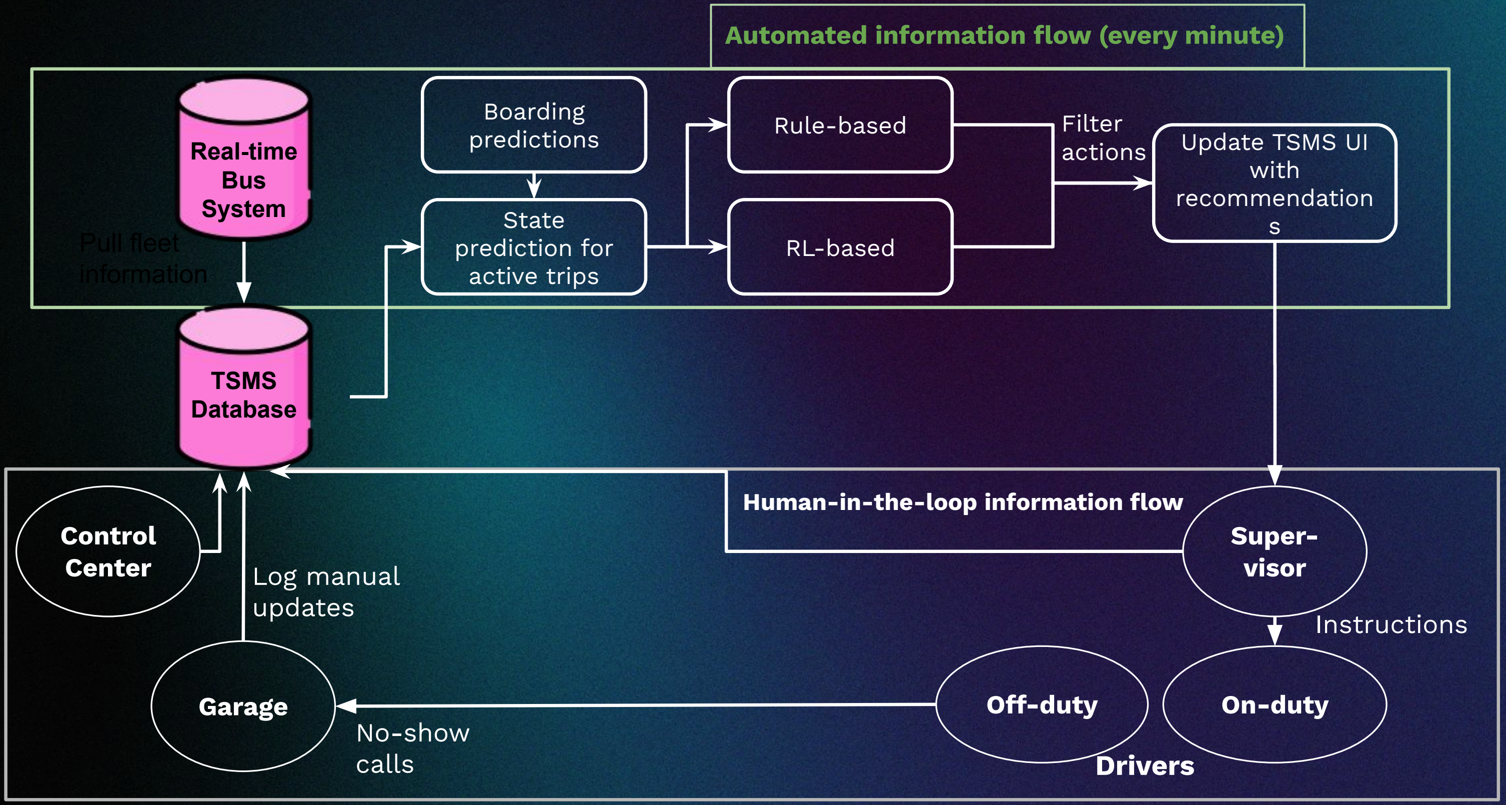


Route:

- ➔ 4 transfers to subway
- ➔ 50 stops each direction
- ➔ AM peak: 6-9 mins
- ➔ 15-25% missing trips
- ➔ 7,500 riders per weekday



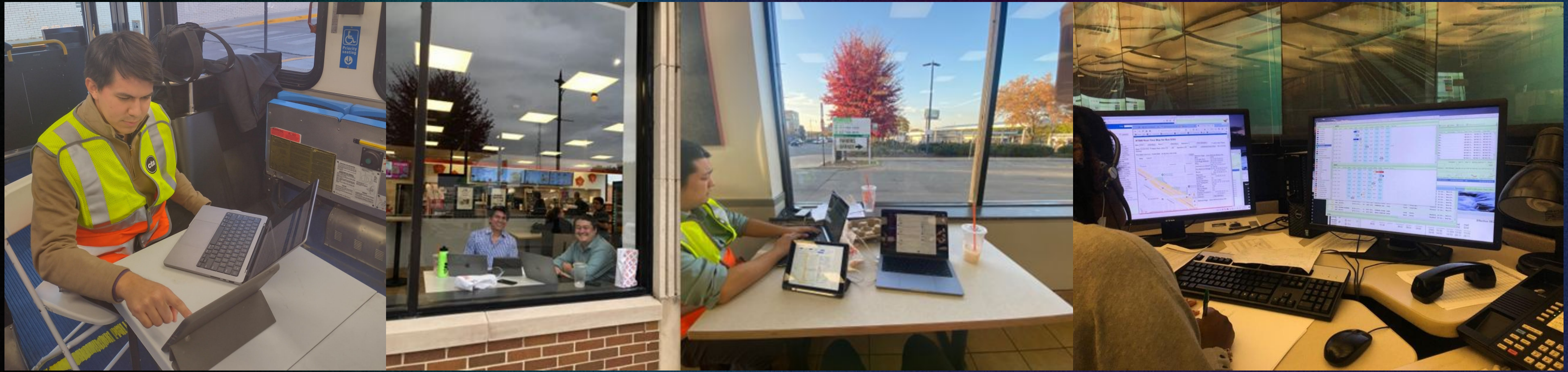
Human in the loop #2: AI recommendation → Human control execution



Human in the loop #3: Field experiment in Chicago

Substantive support from the Chicago Transit Authority (CTA)

- President and CAA
- VP operation
- Garage supervisors
- Drivers



MIT developed the real-time control TSMS database, algorithm, and user interface

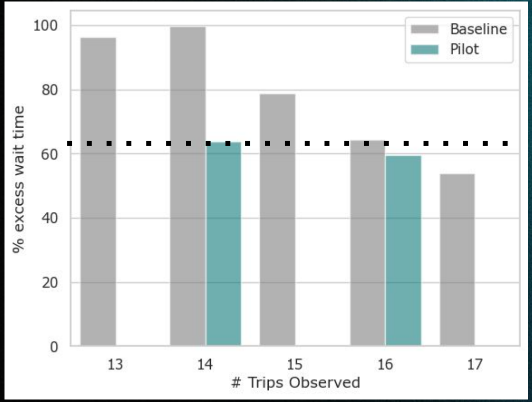
Impacts

Excess Waiting Time: reduced >30%

Impacts of Our Control Strategy: Performance Evaluation

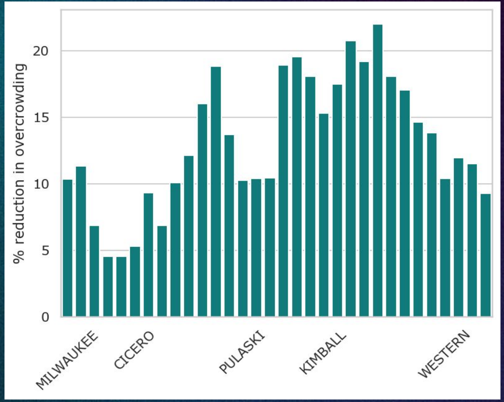
Access wait time by # of bus trips in AM period (6-9am)

37% reduction, equivalent to adding 2 trips



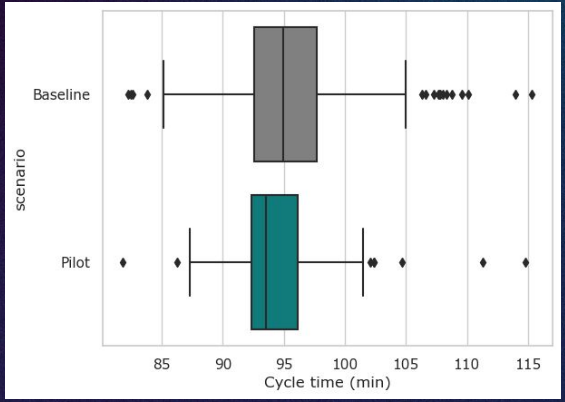
Overcrowding

5-20% reduction



90^{th} Cycle Times

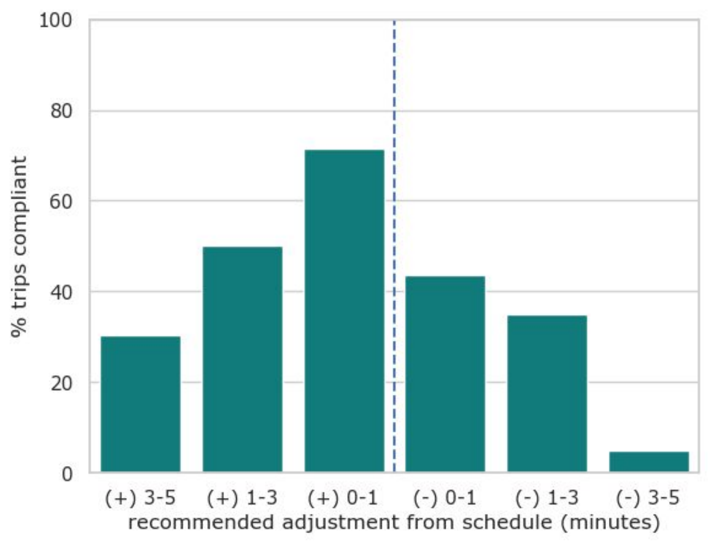
2.5 minutes reduction (2.4%) reduction



Observation #3

Do drivers follow the AI guidance?

Human in the loop #4: Do drivers follow A.I.?



Driver compliance decreases with the aggressiveness of the recommended control actions.

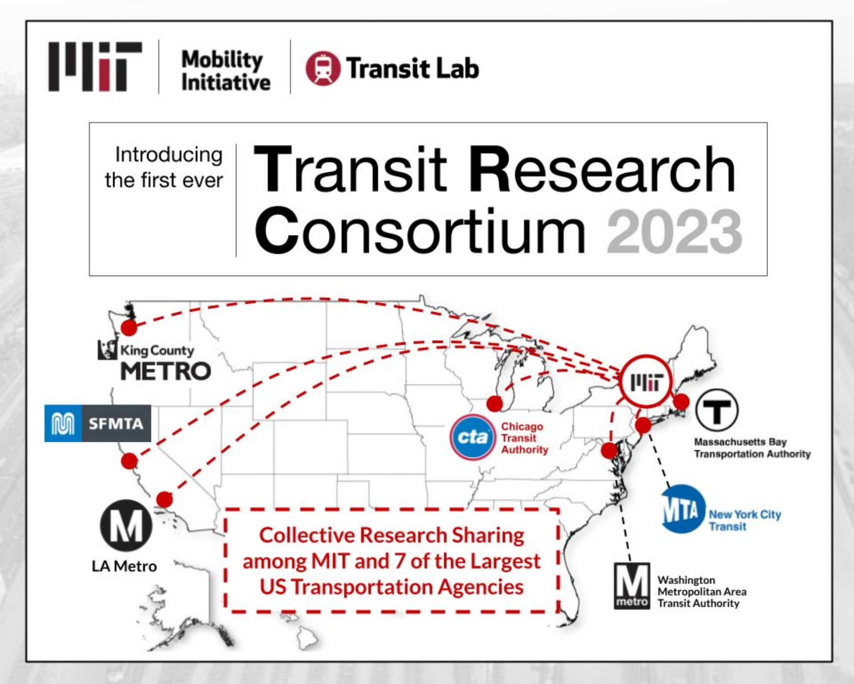
Robustness to uncertainties: Driver compliance

	90th percentile headways	
Baseline (minutes)	Trips with non-compliant drivers	Trips with compliant drivers
27.6	21.8 (-21%)	16.4 (-40%)

Knowledge Transfer: MIT → Chicago

Category	Pilot (BP2)	Full-scale (BP3)
Time period	1 week	1 month
Routes	81	All routes at the Jefferson Park terminal
Control actions	Terminal dispatch	Mid-route holding, dynamic interlining, short-turning
Implementation	RAs overseeing operations	CTA staff takes over the software

Scaling Up: the US transit industry at large



Observation #4

Human in the loop

- Control level
- Garage level
- Organizational level

Represent

Sense

AI for Sentiment Analysis and Complaint Categorization

Michael Leong & Awad Abdelhalim

Data is often unstructured


Tony @_tonymm · Apr 12

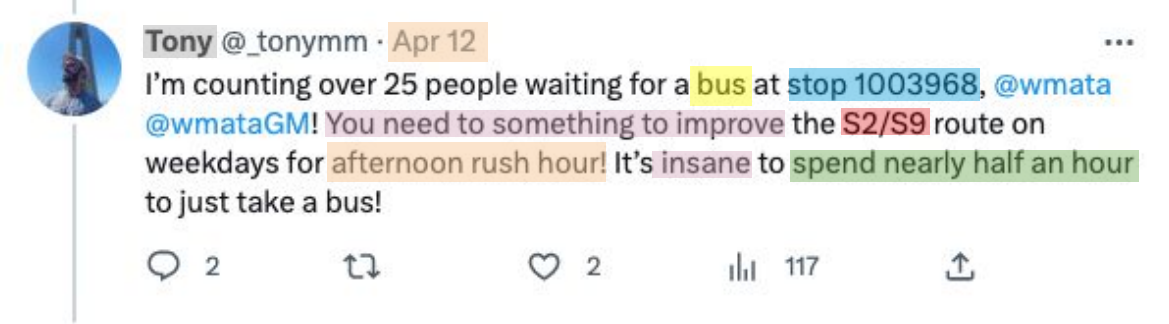
I'm counting over 25 people waiting for a bus at stop 1003968, @wmata @wmataGM! You need to something to improve the S2/S9 route on weekdays for afternoon rush hour! It's insane to spend nearly half an hour to just take a bus!

2 2 117


Jess Anders @jess_manders · Jul 16, 2018

Hey @wmata, nice trash pile to start my morning in car 7354, green line to branch ave. Shame people can't appreciate things but can you send someone to **clean** this up? No one can sit there. Thanks. #metro #wmata

But can provide invaluable insights


Tony @_tonymm · Apr 12
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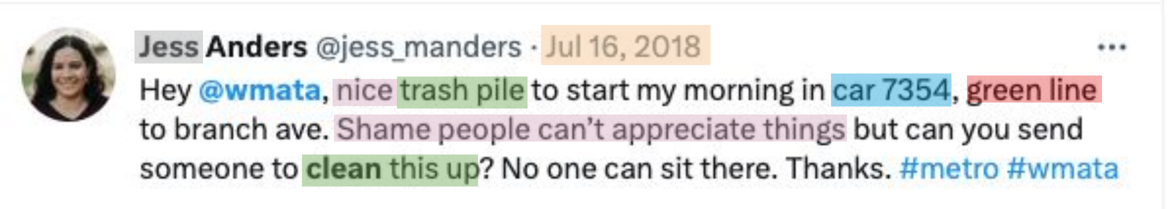
2 2 117

Mode: Bus
Route: S2, S9
Location: Stop 1003968
Time: 4/12/2023 6:48pm
Topic: Schedules / Delays
Sarcasm: Not Present
Sentiment: Negative
Inferred Gender: Male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Mode: Train
Route: Green Line
Vehicle: Car 7354
Time: 7/16/2018 8:03am
Topic: Cleanliness
Sarcasm: Present
Sentiment: Negative
Inferred Gender: Female

Unstructured data → Customer insights

Customer Relationship Management (CRM)
"Comments"

Twitter/X
"Tweets"

Classification

Detected Attributes (Heuristics)

- **Route**
- **Vehicle**
- **Location**
- **Mode**

Inferred Attributes (Models)

- **Topic**
- **Sentiment**
- **Irony**
- **Keywords**
- **Demographics & Profile**

Analysis

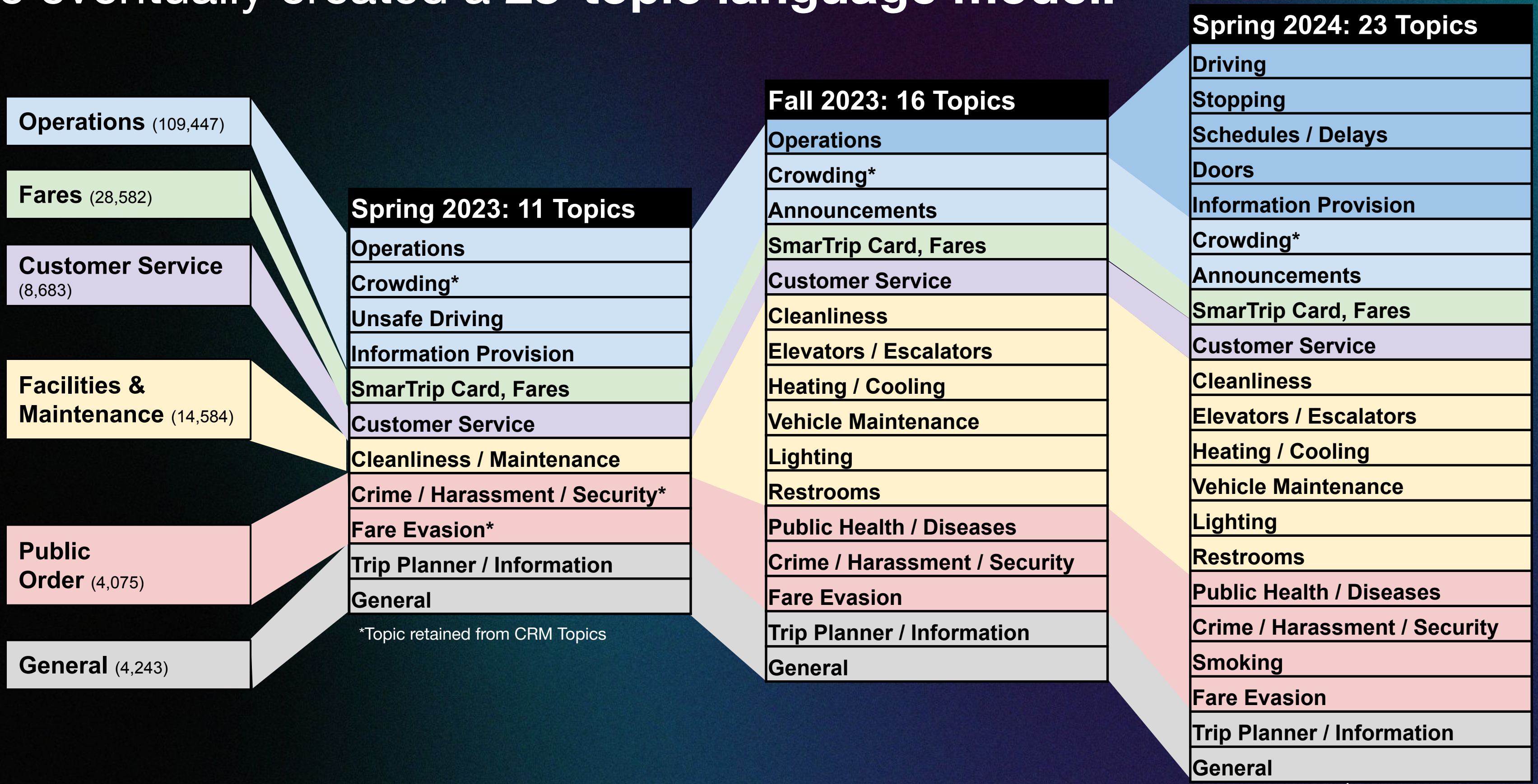
Reporting Metrics

- **Comments per thousand riders**
- **Average Sentiment of Comments**

Use Cases

- **Proactive Monitoring & Detection**
- **Trend & Policy Analysis**
- **Comparative Performance Analysis**

Through further **refinements** and **stakeholder** feedback, we eventually created a **23-topic language model.**



Our current model can **classify any open-text feedback** with an **accuracy of 94%**

Feedback	Prediction
“just want to shout out the station manager at Dupont Circle for helping me out when I had some extenuating circumstances. very personable. an asset to the wmata force for sure. I wish I had gotten his name. Thank You!”	MetRoBERTa Predicted Topic: **Customer Service**
“hey @wmatagm, does @wmata intentionally slow down trains so customers miss connecting buses or is it a byproduct of their complete indifference to customers? train sat outside huntington for several mins just long enough so I could see 161 bus pulling away”	MetRoBERTa Predicted Topic: **Schedules/Delays**
“hey @wmata, bus 7108 is out here on wisconsin in tenleytown just cutting everyone off and going through red lights”	MetRoBERTa Predicted Topic: **Driving**
“one small request @wmata... 8 car trains during rush hour, on the red line to be exact”	MetRoBERTa Predicted Topic: **Crowding**

Our current model can **classify any open-text feedback** with an **accuracy of 94%**

"shoutout to @wmata for these fantastic travel information boards at Metro Center. recently had some relatives in town who took the metro from dulles and to see the monuments all weekend, and they were confident first-time riders because of these investments

MetRoBERTa Predicted Topic:
Information Provision

"@wmata it is sweltering hot inside the Archives station. I don't know how much control you have over this, but a cooler station would make a big difference for customers.

MetRoBERTa Predicted Topic:
Heating/Cooling

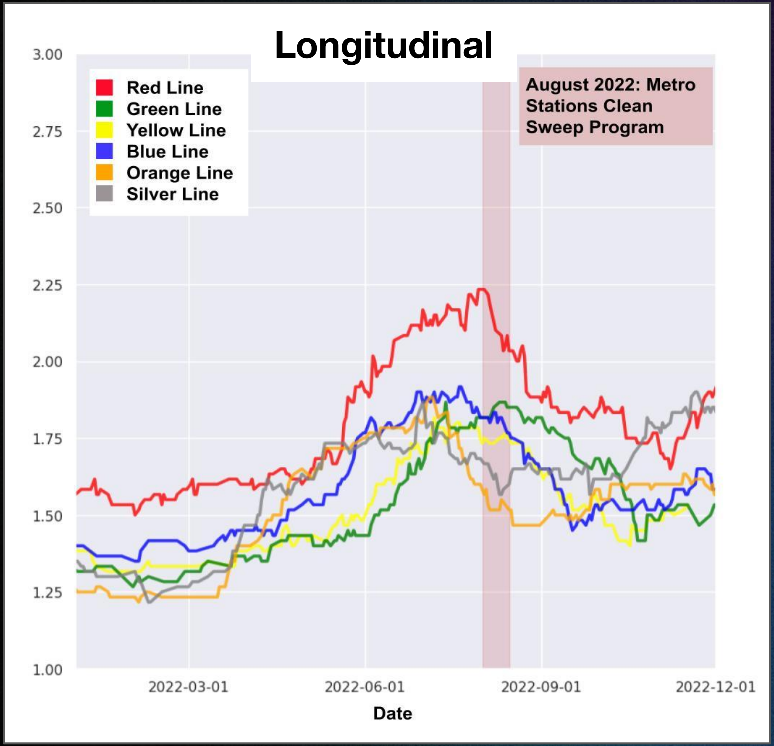
"there is a rat on the south end of the Woodley Park platform. Hanging out under the bench at the end of the platform.

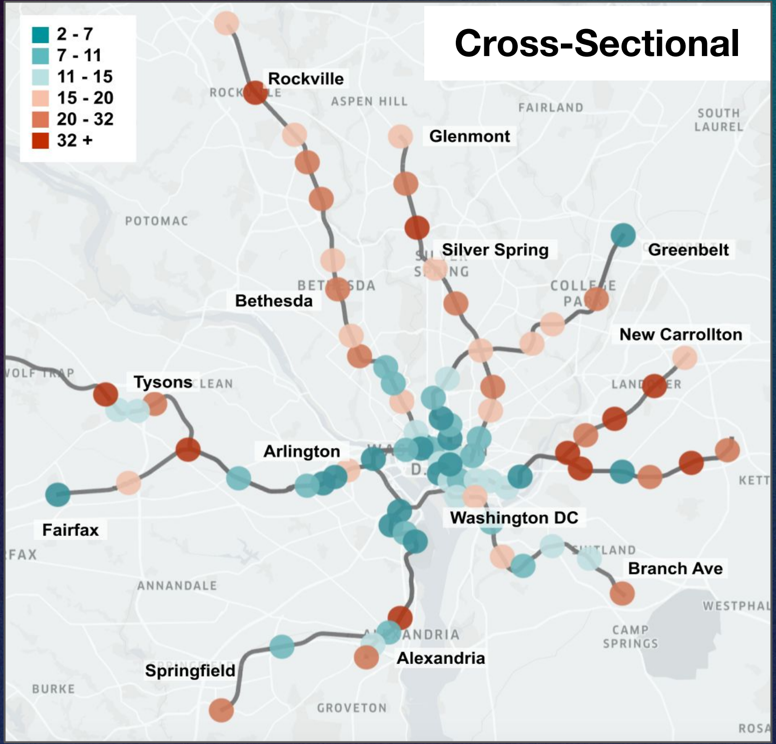
MetRoBERTa Predicted Topic:
Cleanliness

"great smelling elevator that takes me from train platform up to mezzanine to pay and exit the fare gates. thank you!

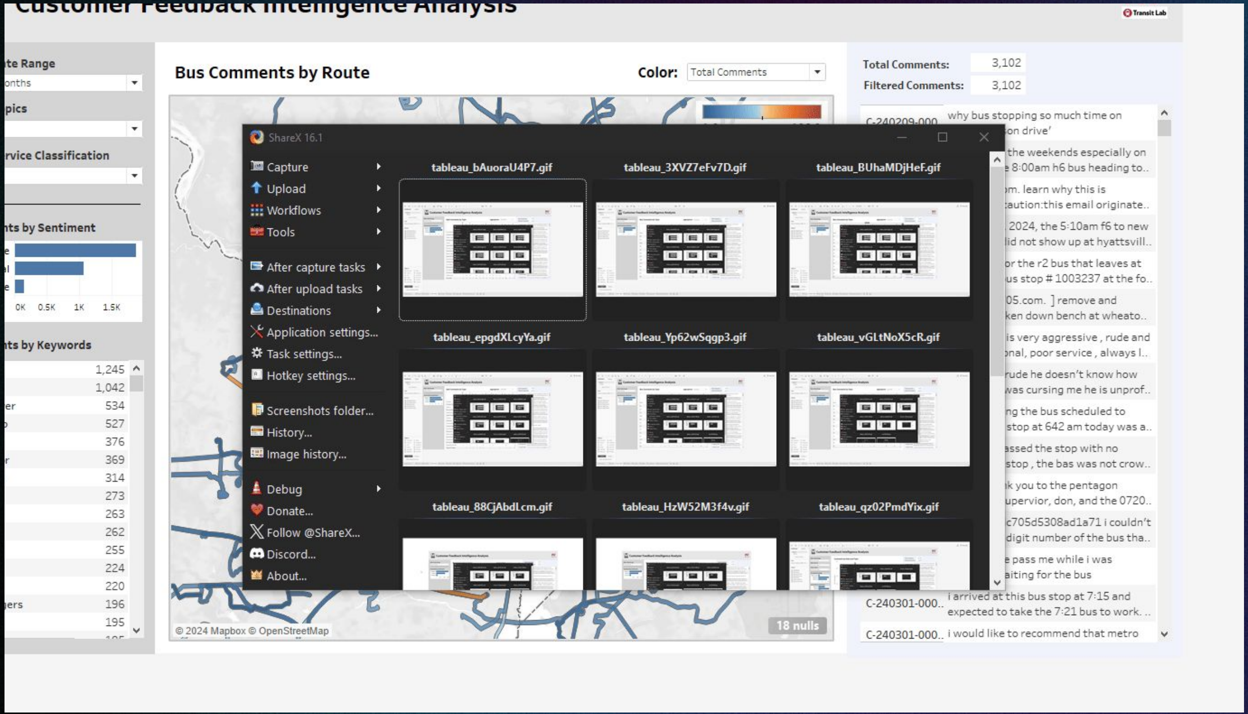
MetRoBERTa Predicted Topic:
Cleanliness

Aggregating & Visualizing Customer Feedback





Customer Feedback Interface deployed at WMATA



Observation #5

What is data?
Respect of qualitative data

MIT/Industry Innovation Pipeline

Industry/MIT Innovation Pipeline

Idea	Method	Industrialization	Implementation
Industry & MIT	MIT	Industry & Consulting Firms	

Monitoring (Industry & MIT)

How do we contribute to the industry?

Solve short-term specific problems

Develop medium-term platforms and capacity

Catalyze strategic institutional changes

Three Levels of Contributions

Specific Requests/Tasks

Stress tests
- Occupying Central
- South Island Line Opening

Disruption analysis
- Signal failure
- Water burst and power outage

Service changes
- 7 → 8 car trains
- Dispatching strategy

Route choice model

Platforms/Capacities

Crowding/Left-behind Estimation

Travel Demand Management

Predictive Analytics

Network Performance Model

Suite of Visualization Toolboxes

Catalyze institutional changes

TfL/MIT Data and Analytics Flow

- AFC
- AVL
- Survey
- Incidence Log
- Wifi
- . . .

→ **TfL Analytics + MIT** →

- London Underground
- London Buses
- London Road
- Fare and Ticketing
- Customer Experience
- TDM
- . . .

MIT Contribution: Results + Methods